# Instructor Run Sheet - REVISED

**PV101 Day 5 - Monday, July 13, 2026**

| **Day 5 focus** | **Commissioning, verification, wire protection, voltage drop, and structured troubleshooting using the systems built on Day 4.** |
|---|---|
| Important change | Do not repeat the small-system build. Preserve the Day 4 builds and use them as live training systems. |

## Learning targets

- Explain voltage, current, resistance, and power in practical language.
- Calculate watts and current for common 12 V loads.
- Identify where fuses belong and explain that they protect conductors.
- Commission an existing PV system using visual inspection and meter evidence.
- Measure voltage drop without rebuilding the system.
- Diagnose safe instructor-created faults using observe, predict, measure, act, verify.
- Leave a system labeled and documented for the next technician.

## Materials

- Five completed Day 4 training systems. Keep them assembled unless a correction is necessary.
- Multimeters, labels/tape, markers, inspection tags, spare fuses, hand tools.
- Optional clamp meter and known 12 V loads.
- Printed quiz, commissioning sheets, fault cards, and exit tickets.

## Schedule

| Time | Block | Instructor notes |
|---|---|---|
| 10:00-10:10 | Landing / readiness | Confirm yesterday's systems are present. No energizing yet. |
| 10:10-10:25 | Morning quiz | Individual. Use results to select review points. |
| 10:25-10:50 | Quiz debrief | Volts, amps, watts, fuse purpose, placement, and voltage drop. |
| 10:50-11:20 | Mini-lesson | Commissioning versus building. A finished system still has to prove itself. |
| 11:20-11:45 | Instructor demonstration | Visual inspection, expected readings, and voltage-drop measurement. |
| 11:45-12:00 | Fault-method briefing | Observe -> predict -> measure -> act -> verify. No random touching. |
| 12:00-12:30 | Lunch / reset | Instructor installs safe faults or prepares fault cards. |
| 12:30-1:15 | Round 1: Commissioning | Teams inspect and document one completed system. |
| 1:15-2:10 | Rounds 2-3: Fault rotation | Teams rotate through safe faults; evidence required before correction. |
| 2:10-2:35 | Wire sizing / voltage-drop application | Use measured or assigned loads to justify conductor and fuse choices. |
| 2:35-3:00 | Teach-back / restore / cleanup | Return every kit to known-good condition and document it. |

## Safe fault menu

- Removed or blown fuse.
- PV disconnected from controller.
- Load disconnected or switched off at an unexpected point.
- Loose, unenergized connection identified before power-up.
- Incorrect meter setting or lead placement as a demonstration fault.
- Mislabeled conductor or missing system label.
- High-resistance connection simulated only with an approved training method; do not create heat intentionally.
- Controller sequence/state issue that can be corrected without bypassing manufacturer instructions.

## Do not use as live faults

- Intentional reverse polarity on energized equipment.
- Unfused battery short circuits.
- Deliberate overheating, damaged conductors, or exposed live copper.
- Any fault that relies on a student “seeing what happens.” We already know what happens.

## Instructor safety holds

- No system energizes until visual inspection and polarity check are complete.
- Any unexpected heat, smell, spark, unstable reading, or damaged insulation means stop and isolate.
- Only one change at a time during troubleshooting.
- After each fault, restore the system and independently verify normal readings.
- The lab win is a defensible diagnosis, not merely making the inverter turn on.